

PUNJAB WHEAT — RAG TRAINING DATASET (2021-2026)

Crop: Wheat | Region: Punjab, Pakistan | Timeline: 2021 to 2026

Purpose: Source corpus for Retrieval-Augmented Generation knowledge base
(chunking ~250-300 words recommended)

HOW TO USE THIS FILE FOR RAG INGESTION

Each numbered section below is a self-contained fact block tagged by **Year**, **Topic**, and **Source** [#]. This structure is designed so that when split into ~300-word chunks, each chunk retains enough context (year + topic) to remain meaningful on its own — important for retrieval quality. Every factual claim carries an inline citation ([#]) matching the master reference list at the end of the file. No claim in this file is uncited.

SECTION A: YEAR-BY-YEAR TIMELINE (2021-2026)

=== YEAR 2021 (Rabi 2020-21 crop / harvested 2021) ===

[A1] **Production baseline entering 2021-22 planning cycle** The Wheat Review Committee (which met in September 2021) noted that Punjab's 2020-21 wheat production stood at 20.9 million tonnes, and directed the province to raise this to 23 million tonnes for the following (2021-22) crop season [25]. Nationally, Pakistan's 2020 wheat output (harvested June 2020) was officially estimated at a below-average 25.25 million tonnes, largely reflecting reduced yields due to heavy rains and hail in March-April 2020 in Punjab province [29].

[A2] **National yield figure, 2020/21 marketing year** USDA FAS data records the national wheat yield for the 2020/21 marketing year at 2.9 tons/hectare [1][23].

[A3] **Sowing-season input crisis heading into the 2021-22 crop** Going into the 2021-22 sowing season, the federal government targeted 30 million tonnes of national wheat production and asked provinces to increase area and per-acre yield using rust-resistant, certified seed, along with proper water, fertilizer, and pesticide usage [25]. However, wheat lost acreage during late sowing when DAP and urea shortages developed, diverting some growers to alternative crops such as potatoes and oilseeds [25].

=== YEAR 2022 (Rabi 2021-22 crop, harvested spring 2022) ===

[A4] **Punjab sowing shortfall for the 2021-22 crop** Punjab, which contributes around 76% of Pakistan's total wheat production, sowed only 16.21 million acres against a target of 16.67

million acres — a shortfall of over 500,000 acres — for the 2021-22 season [25]. Certified seed availability (public and private sector combined) covered only about 25% of requirement, roughly 530,000 tonnes of seed, including 160,000 tonnes of rust-tolerant varieties; the remainder of demand was met from farmers' own retained seed stock [25].

[A5] Minimum support price context, 2021-22 A decision on the minimum support price for the 2021-22 crop was taken late in the season and was set to match Sindh's price of Rs. 2,200 per 40 kg [25]. Over 70% of Punjab's wheat crop that season was reportedly sown within the most suitable window — mid-November to mid-December [25].

[A6] Record yield achieved in South Punjab, 2021-22 The Secretary of Agriculture for South Punjab reported that wheat production in South Punjab increased by an average of 2.73% during 2021-22 despite a 2.41% reduction in cultivated area; regional production rose from 9.239 million metric tons to 9.262 million metric tons. Per-acre yield rose from 34 to 34.92 maunds per acre — described as the first-ever record production achieved in South Punjab [21].

[A7] National area and production, 2021-22 The national wheat-growing area for the 2021-22 season was 8.99 million hectares, with production of 26.39 million tons [14][29]. Separately, the average annual growth in wheat production over the preceding decade (through 2022-23) was calculated at 0.9%, driven mainly by yield improvements of a similar magnitude, with cultivated area remaining largely unchanged [20].

[A8] Prior-season import requirement (context carried into 2022) In the season prior to 2021-22, Pakistan harvested a record 28.75 million tonnes of wheat but still had to import 2.2 million tonnes to meet local requirements and build a 1 million tonne strategic reserve; a 15% yield loss was projected to limit output to no more than 25 million tonnes that year [25]. Trading Corporation of Pakistan (TCP) received import offers between US\$369-383 per tonne (cost and freight included) around this period [25].

[A9] National yield figure, 2021/22 marketing year USDA FAS data records national wheat yield for 2021/22 at 3.0 tons/hectare [1][23].

[A10] The August 2022 floods and their agricultural impact Catastrophic monsoon flooding struck Pakistan from June to October 2022, killing 1,739 people and causing roughly US\$40 billion in damage — described as the deadliest flood event since the 2020 South Asian floods and the worst in Pakistan's history [52]. FAO assessed that approximately 9.4 million acres of crop area nationwide were potentially inundated in August 2022 alone, including 2.7 million acres in Punjab (also 4.8 million acres in Sindh, 1.2 million acres in Balochistan, and 714,000 acres in Khyber Pakhtunkhwa) [50][55]. In Punjab specifically, 223 people died, 3,858 were injured, and roughly 7,200 km² (1.78 million acres) of crops and orchards were affected [52]. More than 1.1 million livestock were killed nationally, including over 205,100 in Punjab [55]. Nationally, the floods destroyed 2.6 million hectares of standing Kharif-season crops across Sindh, Balochistan, and Punjab

[43]. Punjab's flood-exposed crop shares were significant: the province accounts for roughly 16% of national wheat production, 42% of rice, 23% of cotton, and 31% of sugarcane [51].

[A11] Rabi 2022–23 planting despite flood disruption Standing floodwaters reduced planted area for Rabi crops in localized low-lying regions of Sindh and Balochistan, and there were concerns that seed/fertilizer shortages and flood damage to machinery and irrigation infrastructure would further disrupt planting nationally. However, floodwaters receded in time for planting in most areas, and coordinated government and donor efforts ensured adequate seed and fertilizer supply, resulting in near-average plantings for the 2022–23 Rabi season [43].

=== YEAR 2023 (Rabi 2022–23 crop, harvested spring 2023) ===

[A12] National production and target, 2022–23 The 2022–23 wheat production target was 28.37 million tons; actual estimated production came in at 28.16–28.18 million tonnes, up from 26.2 million tonnes the prior year — a 7.45% (or, per a separate estimate, 7.5%) year-on-year increase [19][20]. Sown area increased to 9 million hectares (+0.62%) versus 8.9 million hectares the previous year. The increase was attributed to good weather conditions and timely sowing [19].

[A13] Minimum support prices, 2022–23 Punjab's minimum support price for the 2022–23 crop was set at Rs. 4,000 per 40 kg, while Sindh announced a slightly higher support price of Rs. 4,100 per 40 kg [19].

[A14] Decade-average context (2020–21 to 2022–23) Averaged across 2020–21 through 2022–23, wheat was cultivated on 9.1 million hectares nationally, with average production of about 27.3 million tonnes. The crop accounted for approximately 7.8% of value-added in agriculture and about 1.9% of national GDP [20].

[A15] Domestic price surge, 2022–early 2023 Domestic wheat flour prices rose sharply through 2022 and into early 2023, reaching record highs by June 2023. This price surge was driven by tight market supply, high production and transport costs, and a sharp increase in the minimum support price (MSP) in early 2023. Currency depreciation exceeding 70% between January 2022 and September 2023 made imports significantly more expensive, further amplifying domestic price increases [38].

[A16] Import requirement, 2023–24 marketing year Despite Pakistan's traditional status as a wheat exporter, large quantities were imported over multiple consecutive marketing years to shore up supply and temper prices; for 2023/24 (April–March), wheat imports were forecast at 1.8 million tonnes — above the five-year average but well below the volumes imported between 2020/21 and 2022/23 [38].

[A17] National yield figure, 2022/23 marketing year USDA FAS data records national wheat yield for 2022/23 at 2.9 tons/hectare [1][23].

[A18] Food insecurity context, late 2023 Acute food insecurity (IPC Phase 3/Crisis and above) affected an estimated 11.8 million people between November 2023 and January 2024, up from 10.5 million between April and October 2023 — driven mainly by high food prices eroding purchasing power, compounded by lingering effects of the 2022 floods [38].

=== YEAR 2024 (Rabi 2023–24 crop, harvested spring 2024) ===

[A19] Cost of production detail, 2023–24 crop The Agricultural Policy Institute's cost-of-production survey (conducted May–June 2023 across major wheat-growing areas of Punjab and Sindh) found the market/procurement-center-level cost of production at approximately Rs. 3,303.52 per 40 kg — a 32.73% (Rs. 1,081.33) increase over the prior year, driven primarily by fertilizer/FYM, harvesting/threshing, plant protection, and interculture/ploughing costs [20]. About 87% of national wheat production came from irrigated conditions [20].

[A20] Support price decision, 2023–24 crop The Economic Coordination Committee (ECC) of the cabinet approved a support price of Rs. 3,900 per 40 kg for the 2023–24 wheat crop on 20 December 2023; Sindh subsequently revised its own support price upward to Rs. 4,000 per 40 kg [34]. Farmers receiving the support price during the April–June 2023 post-harvest period were paid Rs. 3,900 per 40 kg equivalent [20].

[A21] Estimated provincial production breakdown, 2023–24 Officials projected Punjab would produce about 22 million tons of wheat for 2023–24, with Sindh expected to reap about 4 million tons, Khyber Pakhtunkhwa 1.4 million tons, and Balochistan 1.3 million tons — implying a national total around 29 million tons, roughly 3 million tons below the 32 million ton target, attributed to elevated input costs (notably a doubling of urea prices due to alleged price manipulation, which prompted a case referred to the Competition Commission of Pakistan) [34].

[A22] National wheat estimate for the 2023 calendar-year cropping season The 2023 cropping season (finalized December 2023) produced an estimated above-average 28.1 million tonnes of wheat nationally, within an overall cereal production total of about 52 million tonnes [38].

[A23] Weather conditions during the 2023–24 growing season Despite warmer-than-average temperatures in December 2023 and January 2024, vegetation conditions were average to above average across main wheat-growing areas due to sufficient irrigation water supply. In minor rainfed ("barani") tracts — including parts of northern Punjab, southwest Balochistan, and southeast Sindh (together about 10% of total wheat sowings) — low rainfall negatively affected crop emergence and early development [37][38].

[A24] National yield figure, 2023/24 marketing year USDA FAS data records national wheat yield for 2023/24 at 3.1 tons/hectare [1][23].

[A25] Record 2024 harvest and area context For the 2024-25 (harvested-2024) season, planted area reached approximately 9 million hectares (some sources cite up to 9.1-9.2 million hectares), up from 8.86 million hectares planted for the 2023 harvest, and slightly higher than the roughly 9 million hectares a year prior [31]. Pakistan achieved a record wheat output of approximately 31.6 million metric tons in the 2024-25 season [8][39] (a separate USDA figure cites 31.8 million tons for the same record season [2]; figures vary slightly by reporting agency and revision cycle).

[A26] Fertilizer market disruption entering the 2024-25 crop Cumulative fertilizer purchases during the 2024 growing season were about 3.2% lower than the previous year. While phosphate and potash sales rose 8.3% and 45.9% respectively, nitrogen (urea) sales fell 6.4% due to price gouging — urea prices reportedly doubled during the season. Persistently high urea prices, combined with dry conditions in December and January, resulted in less-than-ideal fertilizer application at critical points in the crop cycle, likely causing yield penalties [31].

[A27] Support price and market structure change, 2024 Prior to the 2023-24 marketing year, the state-owned Trading Corporation of Pakistan (TCP) purchased most of the country's wheat import requirement. This changed in July 2023 when the government allowed private-sector duty-free wheat imports; private-sector import approvals were reportedly halted as of 31 March 2024, with no clear indication (as of April 2024) that they would resume for 2024-25 [31]. In Punjab, the support price for the 2024 harvest was around Rs. 3,900 per 40 kg (~A\$545/t at the time), while Sindh's was Rs. 4,000 per 40 kg (~US\$559/t); farmers protested, demanding an increase to Rs. 5,000 per 40 kg and a higher procurement target (from 2 million to 5 million tonnes) [31].

[A28] Punjab government exits wheat procurement, 2024 The Punjab provincial government — which had historically purchased wheat from farmers at harvest time at a guaranteed minimum support price — decided not to procure wheat from the 2024 harvest. This absence of the largest traditional buyer increased market supply and put additional downward pressure on prices [37].

[A29] Consumption and per-capita figures, 2024-25 Pakistan's per-capita wheat consumption is around 124 kg per year, among the highest in the world. Population growth combined with continued consumer preference for wheat-based products was projected to push domestic demand 2.3% higher for 2024-25, to 30.9 million tonnes. Total stockfeed-industry use was estimated at only 1.9 million tonnes (6.1% of demand), with the poultry sector accounting for more than 80% of that portion [31]. About 1,000 privately held flour mills produce roughly 40% of flour consumption nationally, with the balance met by at-home processing [3].

[A30] Wheat imports and carry-in stocks, 2024-25 outlook (as forecast in April 2024)

Wheat imports for the 2023-24 season totaled 2.8 million tonnes through the end of February 2024, with a further 500,000 tonnes discharged in March; the government indicated imports would stop at the start of April 2024. USDA/FAS projected 2024-25 carry-in stocks close to a record 4.7 million tonnes (up from 3.9 million tonnes the prior season), well above the government's minimum strategic-reserve requirement of 2.5 million tonnes. Given record production and higher opening stocks, 2024-25 imports were projected to drop significantly, to around 1 million tonnes [31]. A later (2025) USDA update confirmed the 2024/25 wheat production estimate was unchanged at 31.4 million tons at that point in the revision cycle [35].

=== YEAR 2025 (Rabi 2024-25 crop into 2025/26 planning) ===

[A31] Price collapse following the 2024/25 record harvest Wheat prices declined sharply during the 2024/25 harvest season and remained low afterward. The record 2024/25 harvest combined with large import volumes led to high inventories, driving prices down further; the government's decision to abandon its domestic wheat support-price policy exacerbated the decline. Prices stabilized relatively from around August 2024 onward. By March 2025, prices were about 35% lower than the high levels of a year earlier; domestic wheat prices in mid-March 2025 were around US\$245 per ton, with flour prices around US\$317 per ton [3] [36].

[A32] Wheat and flour price trend, August 2024 detail Domestic wheat flour prices declined sharply between March and August 2024 due to abundant market supply from the record 2024 harvest, compounded by the Punjab government's decision not to procure from that harvest. By August 2024, prices averaged 35% below the elevated levels of a year earlier [37].

[A33] Rice and other crop context for the 2024/25 season (cross-reference) Rice production for 2024/25 was adjusted down to 9.76 million tons due to harvested-area revisions, versus an earlier unchanged forecast of a record 10 million tons that had benefited from timely, widespread rains and improved irrigation water availability [35][36].

[A34] Sowing environment entering the 2025/26 crop (autumn 2025) Farmers began planting the 2025/26 crop amid an uncertain policy environment. Government procurement — historically dominated by the Punjab Government — was absent for the 2024/25 crop, and there was no clarity on a government-guaranteed price for the 2025/26 crop, creating uncertainty that directly affected planting decisions. With limited winter-crop alternatives (mainly canola or sesame), most farmers proceeded despite the uncertainty. Monsoon rains in 2025 were adequate to fill Pakistan's two large reservoirs (Mangla and Tarbela), ensuring sufficient irrigation water supply for the wheat crop at sowing time [35].

[A35] Import ban maintained through the 2025/26 marketing year (as of the relevant USDA update) Pakistan had not imported wheat during the then-current marketing year due to a standing import ban; the government indicated it might review the ban later in the marketing year, though imports were considered unlikely. Despite the official export ban on wheat and flour, unofficial exports to Afghanistan were estimated at 500,000 tons (not reflected in official data) [35].

[A36] Wheat sowing progress and Punjab's target, autumn 2025 By mid-November 2025, more than 3 million acres had already been sown in Punjab, as the province worked toward its target of 16.5 million acres for the 2025–26 season. The provincial Agriculture Department reported no shortage of agricultural inputs, with about 100,000 bags of certified wheat seed available through the Punjab Seed Corporation and private-sector companies. Under a post-flood relief package, the provincial government extended interest-free loans to farmers for Rabi crops. Improved soil fertility from recent floods and generally favorable weather were expected to support good yields — echoing the bumper harvest recorded after the 2022 floods [6].

[A37] Recommended high-yield varieties for the 2025–26 Punjab season The Punjab Agriculture Department recommended the following varieties for 2025–26 sowing: Akbar-2019, Dilkash-2020, Fakhar-e-Bhakkar, Aruj-2022, Nishan, Pakistan-2013, Faisalabad-2008, MH-2021, MA-2021, Subhani-2020, and Wafaq-2023 [6].

[A38] Rabi 2025–26 national production target set by the Federal Committee on Agriculture (FCA) The federal government set a wheat production target of about 30 million tonnes for Rabi 2025–26, relying on adequate water availability in Punjab and Sindh. This target followed the Rabi 2024–25 target of 27.92 million tonnes, which was exceeded — actual achieved production was 28.42 million tonnes [10]. (Note: this 28.42 million tonne figure for 2024–25 differs from the ~31.6–31.8 million ton "record" figures cited elsewhere [2][8][39]; the discrepancy likely reflects different measurement bases — target-vs-achieved accounting by the Ministry of National Food Security and Research versus USDA/FAS's independently modeled estimate. Both figures are reported here for transparency, as instructed.)

[A39] Water allocation for Rabi 2025–26 The IRSA Advisory Committee anticipated no water shortage for Punjab and Sindh during Rabi 2025–26, allocating 33.814 million acre-feet (MAF) of water to the two provinces. The Meteorological Department projected below-normal precipitation for most of the country during November–December 2025, with near-normal rainfall expected in January 2026 [10].

[A40] Agricultural credit expansion, FY2025 The State Bank of Pakistan set an indicative agricultural credit disbursement target of Rs. 2.572 trillion for fiscal year 2025 — 16.3% higher than the preceding year's Rs. 2.250 trillion. As of the relevant reporting period, 47 financial institutions were providing agricultural loans, including 5 major commercial banks, 13 medium-sized private banks, 6 Islamic banks, 2 specialized banks (ZTBL and

PPCBL), 11 microfinance banks, and 10 microfinance institutions/rural support programmes [10].

=== YEAR 2026 (Rabi 2025-26 crop, harvested spring 2026 — current season as of this document) ===

[A41] Revised national production estimate, 2025/26 (USDA Grain and Feed Update)

USDA's Grain and Feed Update estimated 2025/26 national wheat production at around 29 million tons — about 8.8% lower than the 2024/25 record of 31.8 million tons. The decline was attributed to a roughly 5.5% decrease in sown area, driven by the absence of a government-guaranteed support price, expectations of low market prices, high sowing-season temperatures, and an extended dry spell [2].

[A42] Earlier, more conservative 2025/26 forecast (USDA Grain and Feed Annual) An earlier USDA Grain and Feed Annual report forecast 2025/26 production at 27.5 million tons — about 13% below the 2024/25 record — citing a 7% decrease in sown area and extremely dry weather. Rainfall was well below average and temperatures significantly above average throughout the growing season [3].

[A43] Punjab-specific production decline, 2025/26 The decline in national wheat production for 2025/26 was most pronounced in Punjab province, which alone reportedly lost around 2 million tons of output relative to the previous year [2].

[A44] Reservoir levels and structural water constraints entering 2025/26 Pakistan's two major reservoirs, Tarbela and Mangla, were running at critically low levels during the 2024/25 growing season, with some reports describing both as at or near "dead level" (unable to release water by gravity flow). Since Punjab and Sindh rely heavily on canal irrigation fed by these reservoirs for their wheat crops, this posed a serious risk. Water demand has grown substantially since the 1970s, while reservoir storage capacity has reportedly declined by about one-third due to sedimentation [8].

[A45] Heat stress and grain-filling impact, 2025/26 Temperatures significantly above average during the 2025/26 season damaged grain-filling at a critical growth stage. Agronomic research indicates that even a 1°C rise above the ideal temperature during grain filling can reduce yield by 5-7%; farmers in central Punjab and upper Sindh experienced this kind of damaging heat during the season [8].

[A46] Absence of a pre-season price signal, 2025/26 For the first time in many years, the government failed to announce a clear wheat support price before the October 2024 start of the 2024-25 sowing season, and this uncertainty persisted into 2025-26 planning; farmers depend on such price signals to guide planting decisions. Without a guaranteed price, many farmers judged the risk too high and reduced wheat area. This shift toward market

deregulation was reportedly influenced partly by IMF reform requirements, removing a traditional safety net for growers [8].

[A47] Consumption, import, and stocks outlook, 2025/26 National wheat consumption for 2025/26 was forecast at around 31.9 million tons, driven mainly by population growth; an estimated 7% of total demand was expected to go to the feed industry, almost entirely to poultry [3]. Given the gap between forecast production and consumption, import requirement estimates for 2025/26 ranged from about 1 million tons (based on the higher, upwardly revised production estimate) [2] to as much as 1.7 million tons (based on more conservative production assumptions) [8]. Wheat ending stocks for 2025/26 were forecast to decline to around 2 million tons, down sharply from the record level of nearly 5 million tons estimated for 2024/25 [2][3].

[A48] Support price and retail price levels, 2025–26 procurement season The government's official wheat support price for Punjab for the 2025–26 procurement season was set at Rs. 3,500 per 40 kg bag. Prices in Sindh and Balochistan tended to run somewhat higher, due to lower local production and higher transport costs. Retail wheat flour (atta) prices ranged between roughly Rs. 4,500 and Rs. 5,837 per 50 kg pack [8].

[A49] Subsidy and relief measures, 2025–26 The Punjab Food Department reportedly supplied wheat to flour mills at subsidized rates to help control atta (flour) prices. Subsidies through the Benazir Income Support Program (BISP) were also under consideration to protect lower-income households from elevated flour costs [8].

[A50] Government-to-government import negotiations, 2025–26 As of early 2026, negotiations were reportedly underway with countries including Russia and Kazakhstan for government-to-government wheat procurement deals at below-market rates, as one option for meeting the projected 2025/26 import requirement without disrupting domestic farmgate prices [8].

[A51] National yield figures, 2024/25 and 2025/26 marketing years USDA FAS data records national wheat yield at 3.2 tons/hectare for both the 2024/25 and 2025/26 marketing years — the highest in the entire 2015–2026 historical series tracked by USDA [1][23].

[A52] 2025 monsoon flooding — context and comparison to 2022 Monsoon flooding in 2025 again affected standing Kharif-season crops in Sindh, Balochistan, and Punjab — the same regions from which the country had not yet fully recovered following the unprecedented 2022 floods. However, the overall flooding extent in 2025 was not as severe as in 2022, particularly in Sindh and Balochistan. Punjab was reported as the most affected province in the 2025 event, recording the worst flooding there in over four decades; roughly 2.9 million people across Punjab were evacuated as of the reporting period, with numbers expected to rise as rains continued [43]. Despite the direct Kharif-season flooding impacts and potential knock-on effects for the Rabi wheat season, adequate wheat reserves carried over from the previous Rabi season, together with improved macroeconomic conditions

(stronger foreign exchange reserves, a stable rupee, and easing food inflation), were expected to help Pakistan absorb the shock without a repeat of 2022-level food security impacts; only short-term, localized, acute food insecurity was anticipated in directly affected areas [43].

SECTION B: FULL NATIONAL YIELD TIME SERIES (USDA FAS, tons/hectare) [1][23]

Marketing Year	Yield (t/ha)
2015/16	2.7
2016/17	2.8
2017/18	3.0
2018/19	2.9
2019/20	2.8
2020/21	2.9
2021/22	3.0
2022/23	2.9
2023/24	3.1
2024/25	3.2
2025/26	3.2

Full dataset source: USDA Foreign Agricultural Service, PSD Online / Wheat Explorer Pakistan [1][23].

SECTION C: WHEAT VARIETIES RELEASED / RECOMMENDED FOR PUNJAB (with release-era context)

[C1] Institutional variety-release history The Wheat Research Institute (WRI) at Ayub Agricultural Research Institute (AARI), Faisalabad has released 65 wheat varieties for various zones since the institute's establishment in 1906 [40][41][42].

[C2] Recent named varieties by release code Confirmed released/recommended varieties include: Anaj-17, Akbar-19 (Akbar-2019), Dilkash-20 (Dilkash-2020), Subhani-21 (Subhani-2020/21), MH-21, Arooj-2022 (also styled Aruj-2022), plus barley varieties (Jau-17, Sultan-17, Talbina-21, Jau-21, Pearl-21) and Durum-21 — Pakistan's first pasta/noodle wheat variety [45].

[C3] MH-21 variety detail MH-21 (varietal code V-12304) is a high-yielding, rust-resistant bread wheat variety for irrigated areas of Punjab. In preliminary (2011–2012) and regular (2012–2013) yield trials it outperformed the check variety Faisalabad-08 by 11.2% and 10.28% respectively in grain yield. It was subsequently tested across 21 locations in Punjab (2013–2014) and 18 locations nationally (2014–2015, 2017–2018, 2019–2020) for yield and disease stability [41][42].

[C4] Dilkash-20 variety detail Dilkash-20 (V-16005) is a newly approved wheat variety for Punjab noted for high yielding potential and disease resistance; its development and multi-location yield trial history (2017 onward, including the Punjab Uniform Wheat Yield Trial and National Uniform Wheat Yield Trials) is documented in dedicated variety-release literature [40].

[C5] Varieties recommended specifically for the 2025–26 Punjab sowing season Akbar-2019, Dilkash-2020, Fakhar-e-Bhakkar, Aruj-2022, Nishan, Pakistan-2013, Faisalabad-2008, MH-2021, MA-2021, Subhani-2020, Wafaq-2023 [6].

[C6] Package-of-Practices (Rabi 2025–26) additional varieties and traits Newer entries in Punjab's official package of practices include specialty end-use types — e.g., a "PBW Biscuit 1 (2024)" premium cookie/biscuit-quality variety (moderately resistant to yellow rust, resistant to brown rust, ~154-day maturity, average yield 17.2 quintals/acre); durum types such as "PDW 291 (2005)" suited for pasta-making, resistant to yellow and brown rust and to loose smut, with low incidence of yellow berry; and general-purpose high-yield lines such as "PBW 872 (2025)" (moderately resistant to yellow and brown rust, ~152-day maturity, bold lustrous grains, average yield 24.4 quintals/acre) and "PBW 824 (2021)" (resistant to brown rust, moderately resistant to yellow rust, ~158-day maturity, average yield 23.2 quintals/acre, recommended for Seeder/Super Seeder use in rice-residue-managed fields) [13].

[C7] Cultivar selection basis used in current agronomic trials Field research (e.g., biochar/irrigation trials at the University of the Punjab, Lahore, 2023–2024) used Dilkash-2020, Akbar-2019, and FSD-08 as representative cultivars, selecting them per AARI's recommended cultivar list for irrigated soils [17][49].

SECTION D: THE 2022 FLOODS — DEDICATED DEEP-DIVE (Cross-Year Reference Topic)

[D1] Scale of the disaster The 2022 Pakistan floods (June–October 2022) killed 1,739 people and caused approximately US\$40 billion in damage, driven by heavier-than-usual monsoon rains and glacier melt following a severe heatwave — both linked to climate change. Pakistan declared a state of emergency on 25 August 2022. It was described as the world's deadliest flood since the 2020 South Asian floods and the worst flood in Pakistan's history [52].

[D2] Crop area affected FAO assessed approximately 9.4 million acres of crop area nationally as potentially inundated in August 2022: 4.8 million acres in Sindh, 2.7 million acres in Punjab, 1.2 million acres in Balochistan, and 714,000 acres in Khyber Pakhtunkhwa [50][55]. At least 1.2 million livestock were killed nationally [50].

[D3] Punjab-specific toll In Punjab, 223 people died and 3,858 were injured; roughly 7,200 km² (1.78 million acres) of crops and orchards were affected. Punjab reported 30% of all recorded injuries nationally [52].

[D4] Sectoral crop damage nationally The floods affected nearly 15% of Pakistan's rice crop and 40% of its cotton crop [50]. Sindh Province alone — the hardest-hit region — accounts for 42% of national rice production, 23% of cotton, and 31% of sugarcane [51][54]. Punjab's exposed shares were 16% of national wheat production, 42% of rice, 23% of cotton, and 31% of sugarcane [51].

[D5] Direct inundation extent (satellite-based) Satellite mapping (Sentinel-1, 22–28 August 2022) showed about 2.5 million hectares of land directly inundated in Sindh alone (over 18% of the province's total area), obstructing access across roughly three-quarters of the province [54][58].

[D6] Impact on Rabi (wheat) planting for 2022–23 Standing floodwaters reduced planted area for Rabi crops in localized low-lying regions of Sindh and Balochistan. There were concerns that seed and fertilizer shortages, plus flood damage to machinery and irrigation infrastructure, would further disrupt planting. However, floodwaters receded in time for planting in most areas, and coordinated government/donor efforts secured adequate seed and fertilizer supplies, resulting in near-average plantings for 2022–23 [43].

[D7] Longer-term food security consequences A UNICEF rapid survey across 15 flood-affected districts (released May 2023) found that one-third of children aged 6–23 months suffered moderate acute malnutrition, with 14% suffering severe acute malnutrition. FAO and WFP's June–November 2023 outlook warned that acute food insecurity was likely to further deteriorate if economic and political conditions worsened, compounding the 2022 flood effects [50].

[D8] Comparison with the 2025 flood event Monsoon flooding recurred in 2025, again affecting Kharif crops in Sindh, Balochistan, and Punjab — regions that had not yet fully recovered from 2022. However, 2025's flooding was less severe overall than 2022, particularly in Sindh and Balochistan, even though Punjab specifically recorded its worst flooding in over four decades during the 2025 event [43].

SECTION E: AGRONOMY AND FIELD-LEVEL PRACTICES (Not Year-Specific — General Reference)

[E1] Botanical classification and cropping system Wheat (*Triticum aestivum*) belongs to the Gramineae family and is grown in all four provinces of Pakistan, particularly Punjab and Sindh, during the winter (Rabi) season [14].

[E2] National average yield by irrigation type Progressive wheat growers in irrigated areas achieve grain yields of 5-7 tonnes/hectare. However, the national average yield is around 3.0 tons/hectare for irrigated wheat and only about 1.2 tons/hectare for rainfed (barani) wheat [24].

[E3] Sowing time criteria Sowing time selection depends on soil and climatic conditions and cropping-zone type. Barani (rainfed) and humid areas require earlier sowing, during October, before the onset of winter [14].

[E4] Recommended sowing windows by province Punjab: 1-20 November (ideal window; official target to complete sowing by 30 November). Sindh: 1-30 November. Khyber Pakhtunkhwa: 15 October-15 November. Late sowing sharply reduces yield — each week of delay past mid-November can cost roughly 10-12 kg of grain per acre due to compressed grain-filling [5].

[E5] Seeding specifications Ideal seed depth is 1.5 cm, with row-to-row spacing of 25 cm and plant-to-plant spacing of 10 cm. Target plant population is 9-10 lac (900,000-1,000,000) per acre, aiming for 30-35 plants per square meter with well-developed, fertile tillers. Standard seed rate is 50-55 kg/acre via seed drill; for late sowing (mid-December to end of December), seed rate increases to 60-70 kg/acre to compensate for reduced tillering under low-temperature conditions [5][14].

[E6] Sowing methods Drill sowing (seed placed at proper depth via seed drill) gives uniform germination and even plant-stand establishment and is preferred over broadcasting. Alternative methods include ridge sowing (recommended for clay loam/heavy loam soils and rainfed areas), furrow-side manual sowing (recommended for saline/waterlogged soils), and zero-tillage direct sowing into rice fields without ploughing (suited to loamy/clayey soils with good drainage), which saves field-preparation time and advances sowing dates [14].

[E7] Micronutrient management — boron Boron deficiency is not typically observed in soils irrigated by canal water with pH below 8, but is a problem in sandy and saline soils. Deficiency disrupts fertilization, reducing grain number and size. Recommended correction: 3–4 kg of boron (gold boron or boro star) per acre at land preparation, or two foliar applications of boron (20% at 125 g/acre, or 5% at 500 ml/acre) 15 days apart [14].

[E8] Micronutrient management — zinc and potash Zinc sulfate (33%) at 6 kg/acre and boric acid (17%) at 2.5 kg/acre are recommended — unless zinc sulfate was already applied to the preceding crop. Potash application increases stalk length, grain number, grain weight, and improves grain color and taste [12][14].

[E9] Fertilizer program by soil fertility class (Punjab Agriculture Department guidance) Weak/poor soils: 2 bags DAP + 2 bags Urea + 1 bag SOP/MOP per acre. Medium fertility soils: proportionately higher DAP, 2 bags Urea, 1 bag SOP/MOP. Fertile soils: 2 bags DAP + 2 bags Urea + 1 bag SOP/MOP per acre (adjusted per soil test). Full phosphorus and potash, plus 1/3 of nitrogen, should be applied at sowing (ideally via band placement with a drill); remaining nitrogen is split across 2–3 later applications. Farmers are urged to have soil tested at government soil and water testing laboratories before finalizing doses [12].

[E10] Irrigation requirement Total crop water requirement is typically 16–18 inches, from a combination of irrigation and rainfall. Pre-sowing irrigation ("rauni") is applied where soil moisture is inadequate to ensure a firm seedbed at depth with fine surface tilth for uniform emergence. Irrigation counts may be reduced on medium-textured soils for certain varieties to prevent lodging [5][11][13].

[E11] Water-saving techniques Laser land leveling and bed planting improve water-use efficiency and reduce weed pressure [11].

[E12] Soil amendment research — biochar A field trial at the Botanical Garden, University of the Punjab, Lahore (2023–2024) tested activated acacia biochar (AAB) at 0, 5, and 10 tons/hectare across three cultivars (Dilkash-2020, Akbar-2019, FSD-08) under five irrigation regimes (100%, 80%, 70%, 60%, 50% of field capacity). The 10-ton AAB treatment improved soil carbon recovery to 45% and carbon sequestration capacity to 49%, versus 0.43% and 0.13% respectively for non-amended soil. Microporosity and water retention improved by 1.1× and 2.2× respectively with 10-ton AAB amendment. Reduced irrigation regimes negatively affected plant growth and yield across all treatments [17][49].

[E13] Agro-ecological zones (rainfall basis) Punjab wheat-growing districts are classified into agro-ecological zones based on average annual rainfall: Cotton zone (~156 mm) — e.g., Khanewal, Bahawalpur; Central mixed zone (~446 mm) — e.g., Faisalabad; Rice zone (~800 mm) — e.g., Gujranwala; Arid zone (~900 mm) — e.g., Chakwal [18].

[E14] Soil and climate parameters Recommended soil types range from sandy loam to clay loam, with ideal pH 6.0–7.5. Optimum sowing temperature is 20–25°C, with cooler temperatures favored during grain-filling [11].

SECTION F: DISEASES AND PESTS (General Reference)

[F1] Major fungal/bacterial diseases Yellow rust (stripe rust), brown rust (leaf rust), stem rust, loose smut, and Karnal bunt are the major wheat diseases in Pakistan [15]. Karnal bunt in particular is addressed in the current Punjab Package of Practices, which notes several newer varieties bred with Karnal bunt resistance [13].

[F2] Disease control strategy Rust diseases (yellow, brown, stem) are managed primarily via resistant/improved varieties. Loose smut and Karnal bunt are controlled through seed treatment with recommended fungicides prior to sowing [15].

[F3] Wheat aphid population dynamics A multi-season field study (2010–2011, ten experimental farms, no pesticide use) tracked wheat aphid populations and their natural enemies across four Punjab agro-ecological zones (cotton zone — Khanewal and Bahawalpur districts; central mixed zone — Faisalabad; rice zone — Gujranwala; arid zone — Chakwal). The commonly grown variety "Sahar" was drill-sown at 125 kg/ha across all sites, October 15–31, in both seasons, following local agronomic recommendations [18].

[F4] Other pest pressures Termites (particularly in sandy/barani soils) and armyworms are additional recurring pest concerns in Pakistani wheat production [11].

SECTION G: RESEARCH INSTITUTIONS AND POLICY BODIES (General Reference)

[G1] Wheat Research Institute (WRI), Faisalabad Formal wheat research in Punjab traces to 1906, with the establishment of a Cereal Section at Lyallpur (now Faisalabad), upgraded to full Wheat Research Institute status in 1975. Subcontinent-wide wheat breeding began around 1907 with a botanical survey identifying 25 landraces across three wheat species; this became the base for subsequent breeding, yielding ten improved varieties released 1911–1965 [16][44].

[G2] Ayub Agricultural Research Institute (AARI) AARI originated in 1962 following the bifurcation of research and education functions from the former Punjab Agricultural College and Research Institute, Lyallpur (established 1906). Its main campus is in Faisalabad, with ecological-zone-specific research institutes, stations, sub-stations, testing centers, service laboratories, and demonstration farms located throughout Punjab. AARI's mission centers on developing new varieties and technologies for food safety, food security, sustainable exportable surplus, value addition, and natural resource conservation [47].

[G3] National production growth since independence Pakistan's national wheat production has grown from about 2.63 million tonnes in 1947–48 to over 25 million tonnes

in recent decades — attributed to the cumulative effect of improved varieties, irrigation expansion, and input intensification [16][44].

[G4] Policy coordination bodies The Federal Committee on Agriculture (FCA) coordinates seasonal production targets, seed availability planning, water allocation (in conjunction with IRSA), and price/support policy across provinces [10]. The Ministry of National Food Security and Research (MNFSR) and the Pakistan Agricultural Research Council (PARC) are additional key national institutions involved in wheat policy and research [9][15].

[G5] Seed production and distribution AARI, Faisalabad provides pre-basic seed of approved varieties (wheat and other crops) at reasonable rates, along with nursery plants for horticultural and citrus crops at subsidized rates [43]. The Punjab Seed Corporation and private-sector companies jointly supply certified seed for the wheat sowing season [6].

SECTION H: PRICES, ECONOMICS, AND FARM-LEVEL FINANCIALS (General / Multi-Year Reference)

[H1] Cost of cultivation range Estimated cost of wheat cultivation per acre falls in a broad range of roughly PKR 32,000–52,000, depending on inputs, region, and farming intensity [4][5]. A separate 2023 API cost-of-production survey found market/procurement-level cost of production at Rs. 3,303.52 per 40 kg for the 2023–24 crop, a 32.73% year-on-year increase [20].

[H2] Profitability and yield per acre One 2026 estimate placed typical net profit at roughly PKR 26,750 per acre under average conditions [4]. Typical yields for a well-managed field are 35–45 maunds per acre, with high-performing farms reaching 50–60 maunds per acre (1 maund $\approx$ 37.3 kg) [5].

[H3] Post-harvest losses Post-harvest losses due to poor threshing and storage are estimated at roughly 10–15% of national production, materially reducing the effective marketable surplus [11].

[H4] Yield gap versus global average Pakistan's average wheat yield (cited at around 1.8 tons/acre in some estimates) still lags the global average of about 2.8 tons/acre, indicating substantial room for improvement via agronomy, mechanization, and input-use efficiency [11].

MASTER REFERENCE LIST (All Sources, [1]–[58] as cited above)

1. USDA Foreign Agricultural Service — *Pakistan Wheat Area, Yield and Production*.
<https://ipad.fas.usda.gov/countrysummary/Default.aspx?id=PK&crop=Wheat>

2. USDA FAS — *Grain and Feed Update, Islamabad, Pakistan* (PK2025-0008).
https://apps.fas.usda.gov/newgainapi/api/Report/DownloadReportByFileName?fileName=Grain+and+Feed+Update_Islamabad_Pakistan_PK2025-0008
3. USDA FAS — *Grain and Feed Annual, Islamabad, Pakistan* (PK2025-0003).
https://apps.fas.usda.gov/newgainapi/api/Report/DownloadReportByFileName?fileName=Grain+and+Feed+Annual_Islamabad_Pakistan_PK2025-0003.pdf
4. IR Farm — *Wheat Farming Profit Per Acre in Pakistan 2026*.
<https://irfarm.com/blogs/latest/wheat-farming-profit-per-acre-pakistan-2026>
5. Times of Agriculture — *Wheat Cultivation in Pakistan 2025: Complete Yield, Cost, and Profit Guide*. <https://www.timesofagriculture.pk/wheat-cultivation-in-pakistan-2025/>
6. UkrAgroConsult — *Pakistan: Wheat sown over 3 mln acres as Punjab targets 16.5 million in 2025-26*. <https://ukragroconsult.com/en/news/pakistan-wheat-sown-over-3-mln-acres-as-punjab-targets-16-5-million-in-2025-26/>
7. FAO GIEWS — *Country Brief: Pakistan* (main page).
<https://www.fao.org/giews/countrybrief/country.jsp?code=PAK>
8. Voyage Freight — *Wheat Harvest 2026 in Pakistan: Production Trends & Price Outlook*. <https://www.voyagefreight.co/supply-chain-management/wheat-harvest-2026-pakistan-production-prices/>
9. Agriculture Department, Government of Punjab (CRS) — *A Detailed Descriptive Study of Wheat Production* (Dr. Qayyum & Pervaiz). [https://crs-agripunjab.punjab.gov.pk/system/files/Wheat%20Descriptive%20Qayyum%20and%20Pervaiz%20\(Dr.%20Qayyum,%20DACRS\).pdf](https://crs-agripunjab.punjab.gov.pk/system/files/Wheat%20Descriptive%20Qayyum%20and%20Pervaiz%20(Dr.%20Qayyum,%20DACRS).pdf)
10. DAWN News — *Rabi wheat target set at 30m tonnes*.
<https://www.dawn.com/news/1951096>
11. Agriculture Pakistan — *Wheat in Pakistan: Yield, Diseases and Fertilizer 2025*.
<https://agriculturepakistan.com/wheat-in-pakistan/>
12. Punjab Portal, Government of the Punjab — *Agriculture Department Punjab Issues Recommendations about Use of Fertilizers for Wheat Cultivation in Irrigated Areas*.
<https://www.punjab.gov.pk/node/5127>
13. Punjab Agricultural University — *Package of Practices for Crops of Punjab, Rabi 2025-26*. https://pau.edu/content/ccil/pf/pp_rabi.pdf
14. GrowTech — *Wheat Production Plan*. <https://www.growtechsol.com/wheat.html>
15. Pakistan Agriculture Research Council (PARC) — *Wheat Program FAQs*.
<http://www.parc.gov.pk/index.php/en/faq-s/60-faq-s/270-faq-s-wheat>
16. Agriculture Department, Government of the Punjab — *Wheat* (AARI crop varieties page). <https://agripunjab.gov.pk/aari-cvwheat>
17. NCBI/PMC — *Optimizing Soil Health through Activated Acacia Biochar under Varying Irrigation Regimes and Cultivars for Sustainable Wheat Cultivation*.
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11745133/>

18. NCBI/PMC — *Spatio-temporal Variations in Wheat Aphid Populations and Their Natural Enemies in Four Agro-ecological Zones of Pakistan*.
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6771227/>
19. Ministry of National Food Security & Research (MNFSR) — *Crops Area & Production (District wise) 2022-23*.
<https://mnfsr.gov.pk/SiteImage/Downloads/Crops%20Area%20AND%20Production%20by%202022-23.pdf>
20. Agricultural Policy Institute (API) — *API Series No. 287: Wheat Policy Analysis for 2023-24 Crop*.
<https://api.gov.pk/SiteImage/Policy/Wheat%20Policy%20Analysis%20For%202023-24%20Crop.pdf>
21. Business Recorder — *'Record wheat yield achieved in South Punjab during 2021-22'*.
<https://www.brecorder.com/news/40179547/record-wheat-yield-achieved-in-south-punjab-during-2021-22>
22. IndexMundi — *Pakistan Wheat Production by Year (1000 MT)*.
<https://www.indexmundi.com/agriculture/?country=pk&commodity=wheat&graph=production>
23. USDA FAS — *Wheat Explorer, Pakistan*.
https://ipad.fas.usda.gov/cropexplorer/cropview/comm_chartview.aspx?ftypeid=47&fattributeid=1&fctypeid=47&fcattributeid=1®ionid=sasia&cntryid=PAK&cropid=0410000&nationalgraph=False&sel_year=2022&startrow=1
24. Pakistan Agricultural Research Council (PARC) — general wheat detail page (progressive vs. national average yields).
<https://www.parc.gov.pk/Detail/MWQ2ZDgzYmEtY2E2Mi00MDU2LWF1MGMtZmUzYjIhZDEzMGEz>
25. DAWN News — *A bleak prognosis of wheat* (Jan 14, 2022).
<https://www.dawn.com/news/1674952>
26. Grain Central — *Pakistan expects record wheat crop*.
<https://www.graincentral.com/markets/pakistan-expects-record-wheat-crop/>
27. The Express Tribune — *Wheat harvest target may be missed* (March 9, 2024).
<https://tribune.com.pk/story/2458822/wheat-harvest-target-may-be-missed>
28. USDA FAS — *Grain and Feed Update, Islamabad, Pakistan* (PK2024-0010).
https://apps.fas.usda.gov/newgainapi/api/Report/DownloadReportByFileName?fileName=Grain+and+Feed+Update_Islamabad_Pakistan_PK2024-0010.pdf
29. FAO GIEWS — *Country Brief Archive, Pakistan*.
https://www.fao.org/giews/countrybrief/country/PAK/pdf_archive/PAK_Archive.pdf
30. ReliefWeb — *GIEWS Country Brief: The Islamic Republic of Pakistan, 05 March 2024*.
<https://reliefweb.int/report/pakistan/giews-country-brief-islamic-republic-pakistan-05-march-2024>

31. Grain Central — *Pakistan expects record wheat crop* (duplicate primary source for A25-A30; see #26).
32. SABRAO Journal of Breeding and Genetics — '*MH-21, A Novel High-Yielding and Rusts Resistant Bread Wheat Variety for Irrigated Areas of Punjab, Pakistan.*
<https://sabraojournal.org/mh-21-novel-high-yielding-rusts-resistant-bread-wheat-variety-irrigated-areas-punjab-pakistan/>
33. SABRAO Journal (PDF) — same study as #32. <https://sabraojournal.org/wp-content/uploads/2023/06/SABRAO-J-Breed-Genet-55-3-749-759-MS22-168.pdf>
34. The Express Tribune — *Wheat harvest target may be missed* (duplicate primary source; see #27).
35. USDA FAS — *Grain and Feed Update* (PK2024-0010) — sowing/policy environment detail (duplicate primary source; see #28).
36. USDA FAS — *Grain and Feed Annual* (PK2025-0003) — price/stocks detail (duplicate primary source; see #3).
37. FAO GIEWS — *Country Brief Archive, Pakistan* — price and weather detail (duplicate primary source; see #29).
38. ReliefWeb — *GIEWS Country Brief, 05 March 2024* — food insecurity and import detail (duplicate primary source; see #30).
39. Voyage Freight — *Wheat Harvest 2026 in Pakistan* — record yield figure (duplicate primary source; see #8).
40. ResearchGate — *DILKASH-20: A Newly Approved Wheat Variety Recommended for Punjab, Pakistan, with Supreme Yielding Potential and Disease Resistance.*
https://www.researchgate.net/publication/370421027_DILKASH-20_A_NEWLY_APPROVED_WHEAT_VARIETY_RECOMMENDED_FOR_PUNJAB_PAKISTAN_WITH_SUPREME_YIELDING_POTENTIAL_AND_DISEASE_RESISTANCE
41. SABRAO Journal — MH-21 variety study (duplicate; see #32).
42. SABRAO Journal (PDF) — MH-21 variety study (duplicate; see #33).
43. GEOGLAM Crop Monitor — *Special Reports (202509).*
<https://www.cropmonitor.org/special-reports-202509>
44. Ayub Agricultural Research Institute — *Wheat* (crop_varieties_wheat page).
https://aari.punjab.gov.pk/crop_varieties_wheat
45. Agriculture Department, Government of the Punjab — *Wheat Research Institute, Faisalabad.* <https://agripunjab.gov.pk/aari-inst-Wheat>
46. Ayub Agricultural Research Institute — *Introduction.*
https://aari.punjab.gov.pk/introductio_n
47. Ayub Agricultural Research Institute — *Welcome to AARI* (homepage).
<https://aari.punjab.gov.pk/>

48. Ayub Agricultural Research Institute — *Wheat Research Institute FAQs*.
https://aari.punjab.gov.pk/faqs_wri
 49. NCBI/PMC — Biochar/irrigation study (duplicate; see #17).
 50. Center for Disaster Philanthropy — *2022 Pakistan Floods*.
<https://disasterphilanthropy.org/disasters/2022-pakistan-floods/>
 51. Nature, Scientific Reports — *A framework for multi-sensor satellite data to evaluate crop production losses: the case study of 2022 Pakistan floods*.
<https://www.nature.com/articles/s41598-023-30347-y>
 52. Wikipedia — *2022 Pakistan floods*. https://en.wikipedia.org/wiki/2022_Pakistan_floods
 53. GEOGLAM Crop Monitor — *Special Reports (202509)* (duplicate; see #43).
 54. ReliefWeb — *The 2022 Pakistan floods: Assessment of crop losses in Sindh Province using satellite data*. <https://reliefweb.int/report/pakistan/2022-pakistan-floods-assessment-crop-losses-sindh-province-using-satellite-data>
 55. Humanitarian Action — *Revised Pakistan 2022 Floods Response Plan: 01 Sep 2022 - 31 May 2023*. <https://humanitarianaction.info/plan/1140/article/revised-pakistan-2022-floods-response-plan-01-sep-2022-31-may-2023-04-oct-2022>
 56. World Bank Group — *Pakistan Floods 2022 Post-Disaster Needs Assessment, Main Report*. <https://thedocs.worldbank.org/en/doc/4a0114eb7d1cecbbbf2f65c5ce0789db-0310012022/original/Pakistan-Floods-2022-PDNA-Main-Report.pdf>
 57. FAO Data in Emergencies — *Pakistan Floods, Post Disaster Needs Assessment (PDNA), 2022*. <https://data-in-emergencies.fao.org/datasets/pakistan-floods-post-disaster-needs-assessment-pdna-2022>
 58. Nature, Scientific Reports / ADS abstract — satellite crop-loss framework (duplicate; see #51).
-

NOTES FOR RAG PIPELINE USE

1. **Chunking recommendation:** Each lettered/numbered fact block (e.g., [A1], [E5], [D3]) is already sized close to your target ~300-word chunk. Consider chunking at this block level rather than by raw word-count, to preserve the year/topic tag as chunk metadata.
2. **Metadata tagging suggestion:** Tag each chunk with `(crop=wheat)`, `(region=punjab)`, `(year=<YYYY>)` (from the block's section header), and `(topic=<production|price|policy|variety|disease|flood|agronomy>)` for more precise retrieval filtering.
3. **Conflicting figures flagged intentionally:** Where sources disagree (e.g., 2024/25 production cited as 31.4M, 31.6M, 31.8M, or 28.42M tons depending on agency and vintage), this file preserves all figures with their source rather than silently picking

one — your retrieval/generation prompt should instruct the LLM to note such discrepancies rather than presenting a single number as certain.

4. **Duplicate reference entries:** Several reference numbers point to the same underlying document (noted as "duplicate primary source" in the list) because the same source was cited across multiple, non-adjacent fact blocks during compilation. Deduplicate URLs during ingestion if your pipeline requires unique source IDs.